



INTERNATIONAL TELECOMMUNICATION UNION

**TELECOMMUNICATION
STANDARDIZATION SECTOR**

STUDY PERIOD 2022-2024

SG2-Cn

STUDY GROUP 2

Original: English

Question(s): 1/2

Geneva, 08-17 Nov 2023

CONTRIBUTION

Source: Ministry of Communications, India

Title: TR.MMWF: Proposed modifications

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Abstract: This contribution attempts to provide an updated draft text for draft recommendation ITU-T TR.MMWF- technical report on Methodologies to mitigate Wangiri Fraud at Annexure I. The reference text is [SG2-TD229-R1/PLEN \(2023-03\)](#)

Field Code Changed

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Annex 1

I n t e r n a t i o n a l T e l e c o m m u n i c a t i o n U n i o n

ITU-T Technical Report

TELECOMMUNICATION
STANDARDIZATION SECTOR
OF ITU

(22 March 2023)

Methodologies to mitigate Wangiri Fraud

ITU-T



Summary

This technical report aims to define Wangiri fraud in telecommunications and to provide the most effective methodologies to mitigate this type of fraud

Keywords

[Wangiri; Premium rated numbers; Fraud]

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Technical Report ITU-T Methodologies to mitigate Wangiri Fraud

Summary

This technical report aims to define Wangiri fraud in telecommunications and to provide the most effective methodologies to mitigate this type of fraud.

1 Scope

This technical report provides a definition of Wangiri fraud and **some** methodologies to **detect and** mitigate ~~of~~ this type of telecommunications fraud. Moreover, this technical report will describe the possible role of Member State **and Recognized Operating Agencies ROAs**/Service Providers/Manufactures/**Regulators** to fight ~~this against the Wangiri~~ fraud globally.

2 References

The following ITU-T Recommendations and other references contain provisions which, through reference in this text, constitute provisions of this Technical Report. At the time of publication, the editions indicated were valid. All Recommendations and other references are subject to revision; users of this Technical Report are therefore encouraged to investigate the possibility of applying the most recent edition of the Recommendations and other references listed below. A list of the currently valid ITU-T Recommendations is regularly published. The reference to a document within this Technical Report does not give it, as a stand-alone document.

- [1] [ITU-T E.101] Recommendation ITU-T E.101 (2009), *Definitions of terms used for identifiers (names, numbers, addresses and other identifiers) for public telecommunication services and networks in the E-series Recommendations*.
- [2] [ITU-T E.157] Recommendation ITU-T E.157 (2021), *International calling party number delivery*
- [3] [ITU-T E.164] Recommendation ITU-T E.164 (2010), *The international public telecommunication numbering plan*.

3 Terms and definitions

3.1 Terms defined elsewhere

This Technical Report uses the following terms defined elsewhere:

- 3.1.1. **Country** [ITU-T E.101]: A specific country, a group of countries in an integrated numbering plan or a specific geographical area.
- 3.1.2. **Operator** [ITU-T E.101]: An operating agency providing public telecommunication networks or public telecommunication services.
- 3.1.3. **National destination code (NDC)** **[ITU-T E.101]**: A nationally optional code field, within the international public telecommunication numbering plan (hereafter referred to as the "international E.164-numbering plan"), which – combined with the subscriber's number (SN) – will constitute the national (significant) number of the international E.164-number for geographic areas.
- 3.1.4. **calling party number [ITU-T E.157]**: The ITU-T E.164 number of the originator of the call or a special allocated number.
- 3.1.5. **international calling party number delivery [ITU-T E.157]**: Calling party number delivery across boundaries of countries.

3.1.6. **special allocated number [ITU-T E.157]**: A number from a series that the numbering plan administrator (NPA)/national regulatory authority (NRA) has allocated for the specific purpose of enabling operators/providers to show that they are responsible for a call when the original calling party number (CPN) cannot be transmitted.

3.1.7.

3.2. Terms defined here

This Technical Report defines the following terms:

3.2.1. **Wangiri**: type of telecom fraud ~~when~~ using short or one ring calls (“one ring and drop”) to cheat subscribers by prompting them to call back the number, which is a Premium Rate Number (PRN), a premium rate number with resulting in the victim incurring very expensive charges/ rates.

3.2.2. **Premium Rate Numbers**: Special numbers dedicated to a certain service with ~~declared~~ charged rate higher than usual rate.

4 Abbreviations

CLI	Calling Line Identity
CPN	Calling Party Number
CPND	Calling Party Number Delivery
PRN	Premium Rate Numbers
PRS	Premium Rate Service
IPRN	<u>International Premium Rate Numbers</u>
IVR	Interactive Voice response
NDC	National Destination Code
PSTN	Public Switched Telephone Network
PLMN	Public Land Mobile Network
CDRs	Call Detail Records
ROAs	Recognized Operating Agencies

5 Introduction

The rapid growth of the telecommunication landscape leads to a rapid rise of frauds using Telecom infrastructure. Wangiri is a common fraud activity targeting subscribers as well as telecom operators, in this type of fraud which, fraudsters make calls but hang up after the first ring to entice/ prompt victims to call back. Hence, the name Wangiri, which in Japanese word means “one (ring) and cut.” When victims call back, they don’t realise that it’s a premium rated number, and they’re hit with very high charges without their knowledge. This problem is growing rapidly in different countries but in general cases these premium [rate] numbers seems to be from developing countries.

A large volume of calls are delivered to the target networks with no apparent intention of establishing a conversational call setup (zero duration calls). [Also often, no CDRs are generated in many management and signalling systems because the calls are not answered (missed calls only)].

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6 Wangiri Definition

'Wangiri' is a Japanese word, which literally translates as 'one ring and drop'. And a Wangiri call does just that. Calls usually cut off just as the phone rings, leaving a missed call message from an international, or unusual number. When the unsuspecting victim returns the call, they are routed to an expensive, premium rate service that earns the scammer a fortune in commissions.

In Wangiri fraud the scammers generate single beep phone calls to many phone users using an auto-dialer. When the user calls back out of curiosity on receiving a missed call, the scammers try to keep the victims busy on the line for as long as possible. The calls are directed to either a real person or an Interactive Voice Response (IVR) system to extract as much money as possible since the call is routed to a premium-rate number.

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7 Background and description of the fraud

Telecom operators are facing this type of fraud since 2009 when the earlier cases found in Japan, from that time this type of fraud is growing exponentially year by year all over the world. In Wangiri, a fraudster gives a missed call to several victim's phone numbers of different countries from an international or unusual number. To the subscriber, the CPN/CLI is modified in such a way that it looks like a genuine call and when the victim calls back, it turns out to be a premium rate number (PRN) number owned by fraudster for which the victim is charged heavily for the calls. The fraudster intends to keep the victim on hold to increase the billed amount. The premium rate service provider pays the fraudster a certain share of the call revenue for each minute of call received by the premium rate number.

As declared in multiple fraud loss survey reports, Wangiri is one of the top 5 fraud methods used by fraudsters to carry out fraudulent activities

7.1 Impact of Wangiri fraud on telecom operators and customer: Some Telecom operators suffer from both direct and in-direct loss due to Wangiri Fraud. Wangiri impacts the customers adversely resulting in customer churn due to high customer dissatisfaction from bill shocks and bad customer experience. It also has a negative impact on the operator brand image as subscribers would complain about the high call rates charged to them without them being aware of it. There is also an impact on the cost due to the increase in customer management.

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Why Wangiri cannot be eliminated completely? ~~The main~~

7.2 Reasons due to which Wangiri cannot be eliminated but can be managed are:

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- There is no proper rule or regulation defined on the carrier business causing the fraudster to take advantage of the vulnerabilities of the carrier network.
- Some operators do not take action on Wangiri fraud, as it relates to large volumes of in/out calls and thereby generates significant revenues;
- Limited implementation of E.157 Calling Party Number Delivery (CPND) requirements through the whole call route path.
- Lack of strict regulation on the CPND The lack of visibility on the end carrier who is terminating the call is another reason why Wangiri can't be eliminated.
- Most Wangiri frauds are conducted using international numbers. Telecom operators can't effectively track them due to the complications concerning international calls.
- Wangiri fraudsters keep changing their fraudulent behavior continuously
- Numbering range for PRN/PRS looks similar to ordinary phone numbers
- Limited/light possibilities for CDR in/out monitoring and management
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The end carrier who terminates a fraudulent missed call is not aware of the fraudster details and the country from which call has originated is a high-risk destination or not. Enhanced awareness at the carrier level can significantly decrease many Wangiri cases.

8 Proposed Methodologies to mitigate Wangiri fraud

Detect Wangiri fraud attacks as quickly as possible and minimize the fraud window during which most of the losses occur. In the following Some methodologies to reduce Wangiri risk are described below. All Member States and Recognized Operating Agencies ROAs/Service providers/Manufactures/Regulators are invited to consider the following measures to mitigate the Wangiri fraud:

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8.1. Routing Management of Carrier: When a fraudster carries out the Wangiri fraud and gives missed calls to multiple subscribers, the augmented amount of traffic can be perceived by the carrier who routes these calls. If an operator examines this activity, there will be a rhythmic trend of increased traffic observed on the same carrier to route these calls. In such cases, an operator might take necessary protections to route all the traffic through an alternative carrier. Routing the calls through a different carrier will help in breaking the chain between the fraudster and the linked carrier. An appropriate regulations of international traffic by member states might help in relieving the problem to a greater extent.

8.2. Customer Experience Management: As the customer is the main target of any business, telecom operators need to manage their customer complaints efficiently, which will reduce customer churn and also loss of revenue. All employees in telecom operator could be informed about Wangiri and how to manage complaints related to this fraud.

8.3. Subscriber Awareness: A pro-active approach to minimize Wangiri fraud would involve making the subscribers consumers aware of the risk. Customers may be alerted through Information, Education and Communication (IEC) campaign by the operators about the risk involved in calling back missed calls received from International Numbers. There are several free apps which have active community platforms where users continuously report spam numbers. These apps which can be used to quickly check suspicious numbers. This would tell the customer if the number is a part of any ongoing scams or not and will help them in blocking the spam numbers used for Wangiri calls.

8.4. Interactive Voice Response (IVR) Facility: Whenever a subscriber calls back to a subspecies destination upon receiving a missed call, the telecom operator might have an IVR voice informing him about his called destination. This IVR warning would help the subscriber to decide if they really want to make the call or not. through the IVR announcement that the number you are calling is a premium rate number and involves high calling charges.

8.5. Using Artificial Intelligence & Machine Learning: Telecom Operators might have an Artificial Intelligence and Machine Learning system to analyze call data and identify anomalies associated with Wangiri fraud and to make alerts to the customers via different means when receive missed calls from subspecies numbers.

8.6. Distributed Ledger Technology (DLT) Platform: DLT, among other solutions, Telecom operators may use operator-run portals such as DLT platforms where all the operators present in the panel will be inter connected with each other. They can login and share intelligence in near to real-time about the actual Wangiri calls that have already occurred. This data can help the telecom operators in the proactive management of future wangri calls, increasing the likelihood of correctly identifying a Wangiri call before the intended victim suffers harm and prevent the fraudulent practices.

8.7. Numbers Blocking: Implement real-time call monitoring and analysis to detect patterns indicative of Wangiri fraud, such as a high volume of short-duration calls. Regulators in different countries might allow telecom operators to block all numbers that were repeated in Wangiri fraud.

8.8. Information, Education and Communication(IEC) campaign by the regulators is very much essential for dissemination of information about the charging plan of Premium Rate

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numbers because a large segment of customers are not aware about the high rates of such premium rate numbers.

8.9. **Sharing of Black Lists:** Regulators in different countries might set up a mechanism to share the most frequent numbers committing Wangiri fraud.

8.10. **The proper National Numbering Plans management:** Regulators might prepare a mechanism to ensure the proper use of NDCs with operators in order to avoid different types of fraud including Wangiri, call masking and spoofing.

8.11. **The Application of WTSA Res. 91:** All member states are invited by this Resolution to “pursuant to the relevant ITU T Recommendations, to make available information on the presentation of their national numbering plans and amendments thereto in a timely manner, so as to ensure that the electronic repository remains up to date.”

8.12. Regulators should assign short code PRN, different from ordinary operators range

8.13. Publishing of Premium Rate Numbers: Regulators may mandate telecom operators to publish the premium rate numbers, accessible from their network along with charging rates, on their website or in common national or international database.

8.14. Publishing of PRN numbering ranges: Regulator may publish PRN in official national and international information resources.

8.15. Fixing Ceiling rates: In order to protect interests of end subscribers and telecom operators, regulators may evaluate fixing of some ceiling rates which can be applied for providing premium rate services using PSTN and PLMN.

8.16. Strong Implementation of Recommendation ITU-T E.157 and relevant ITU-T Recommendations: this methodology is helping to mitigate telecommunications fraud in general not only Wangiri fraud.

8.17. Advanced telecommunication management and signaling systems: Manufacturer may implement short call CDR detection in every telecommunication management and signaling system.

8.18. Agreements between networks, that foresee monitoring high traffic peaks with unanswered calls and tracking calls from originating network to destination.

8.19. **Removal of International Services as the default service for a subscriber:** International service calling facility should not be activated on prepaid SIM cards without the explicit approval of the consumer.

8.20. Regulatory Cooperation: Work with regulatory authorities and other telecom providers to share information about known Wangiri numbers and collaborate on fraud prevention efforts.

8.18-8.21. **Geolocation Filtering:** Implement geolocation filtering to block incoming calls from specific regions known to be sources of Wangiri fraud.

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